

Alexander Prospal

Cleveland, OH

Phone: (216) 339-1306 — Email: alexander@prospal.dev
GitHub: github.com/AlexProspal — Website: prospal.dev

Education

Bachelor of Science in Computer Science (Math Minor) Expected Graduation: May 2028
Case Western Reserve University, Cleveland, OH

Bachelor of Arts in Economics (Quantitative Methods Track) Expected Graduation: May 2028
Case Western Reserve University, Cleveland, OH

Projects

CIFAR-10 Residual Convolutional Neural Network CSDS 465: Computer Vision

- Designed and trained a 3.97M-parameter residual CNN in PyTorch for 10-class CIFAR-10 image classification, achieving 93.12% test accuracy
- Built a training pipeline using residual blocks, batch normalization, dropout, data augmentation, label smoothing, Adam optimization, weight decay, and scheduled learning-rate reduction
- Implemented reusable training and inference modules with model checkpointing, single-image classification, and visualization of all 32 first-layer convolutional activation maps

SpartieSwap — Campus Borrowing & Lending Platform CSDS 393: Software Engineering

- Built a campus-only borrowing and lending web application with a four-person Scrum team using Django, Python, PostgreSQL, Bootstrap, and Docker
- Developed the complete exchange workflow including listings, search and filtering, borrowing requests, approvals, pickup and return tracking, histories, overdue indicators, and two-way reviews
- Integrated server-side authorization, relational business rules, and GitHub Actions CI with PostgreSQL, Ruff, migrations, and automated testing; delivered all 10 user stories and 40 story points with 153 passing tests and 98% measured code coverage

Secure Password Manager Application CSDS 344: Computer Security

- Designed and implemented a secure password manager in Java using MVC architecture, encrypted persistent storage, authentication, and controlled credential access
- Built a JavaFX interface supporting credential creation, retrieval, modification, and deletion with input validation and separation of security-sensitive application logic

Theater Inventory Database System CSDS 341: Introduction to Database Systems

- Designed a normalized PostgreSQL database for managing costumes, props, performers, inventory availability, and production assignments using primary, foreign, and composite keys
- Developed SQL queries using joins, aggregation, nested subqueries, and constraint-based filtering alongside a Python command-line application with parameterized queries and inventory business logic

Technical Skills

- **Languages:** Python, Java, SQL, JavaScript, R, HTML, CSS
- **Frameworks & Libraries:** PyTorch, Torchvision, Django, JavaFX, Flask, Bootstrap, scikit-learn, Pandas, NumPy, Matplotlib
- **Databases & Development Tools:** PostgreSQL, Git, GitHub, GitHub Actions, Docker, Linux, Ruff, SLURM
- **Technical Areas:** Computer Vision, Image Processing, Deep Learning, Machine Learning, Software Engineering, Algorithms, Database Design, Secure Software Development, HPC/Cluster Computing

Relevant Coursework

Current – Fall 2026: CSDS 423 Numerical Algorithms for Machine Learning, CSDS 465 Computer Vision, CSDS 490 Digital Image Processing

Completed: Introduction to Machine Learning, Algorithms, Software Engineering, Introduction to Database Systems, Computer Security, Data Visualization

Mathematics: Linear Algebra, Calculus III, Statistics, Discrete Mathematics